Mathematical and Computational Methods

Exercise Sheet 6: Eigensystems

Tutorial route (▶, in this order, about 75 minutes): 1.5, 2.2, 2.4, 2.5, 2.7, 2.10. Your tutor works 2.3 at the board before 2.7. Everything else is for home study.
Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 *(⋆ drill – the eigen-relation).* Let $\mathbf{A} = \begin{pmatrix} 4 & 2 \\ 1 & 3 \end{pmatrix}$. Verify that $\mathbf{u} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ are eigenvectors, and state the eigenvalue of each. Check your two eigenvalues against $\operatorname{tr}\mathbf{A}$ and $\det\mathbf{A}$.

Exercise 1.2 *(⋆ drill – eigenvalues and eigenvectors of a 2×2).* Let $\mathbf{A} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$. (a) Write down the characteristic equation and solve it for the eigenvalues. (b) Find an eigenvector for each eigenvalue.

Exercise 1.3 *(⋆ drill – complex eigenvalues).* (a) Find the eigenvalues of $\mathbf{R} = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$, a quarter-turn that also doubles lengths. (b) Using $\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$, write down the eigenvalues of the 60° rotation.

Exercise 1.4 *(⋆ drill – spot the structure; the conjugate product).* (a) For each matrix say whether it is *symmetric*, *Hermitian*, both, or neither:

$$\text{(i)}\ \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix}, \quad \text{(ii)}\ \begin{pmatrix} 2 & \mathrm{i} \\ -\mathrm{i} & 3 \end{pmatrix}, \quad \text{(iii)}\ \begin{pmatrix} 0 & 3 \\ -3 & 0 \end{pmatrix}, \quad \text{(iv)}\ \begin{pmatrix} 1 & 2+\mathrm{i} \\ 2+\mathrm{i} & 4 \end{pmatrix}.$$

(b) For $\mathbf{u} = (1, \mathrm{i})^{\mathsf{T}}$ and $\mathbf{w} = (\mathrm{i}, 1)^{\mathsf{T}}$ compute the conjugate inner product $\mathbf{u}^{\dagger}\mathbf{w} = \overline{u_1}w_1 + \overline{u_2}w_2$ and the plain product $\mathbf{u}^{\mathsf{T}}\mathbf{w}$. Are $\mathbf{u}$ and $\mathbf{w}$ orthogonal?

Exercise 1.5 *(▶ ⋆ drill – diagonalisable?).* Decide, with a one-line reason, whether each matrix is diagonalisable:

$$\text{(a)}\ \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}, \qquad \text{(b)}\ \begin{pmatrix} 3 & 1 \\ 0 & 3 \end{pmatrix}, \qquad \text{(c)}\ \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}.$$

Exercise 1.6 *(⋆ drill – assembling $\mathbf{P}$ and $\mathbf{D}$).* A 2×2 matrix $\mathbf{A}$ has eigenpairs $\lambda_1 = 3$, $\mathbf{v}_1 = (1,1)^{\mathsf{T}}$ and $\lambda_2 = -2$, $\mathbf{v}_2 = (1,-1)^{\mathsf{T}}$. (a) Write down matrices $\mathbf{P}$ and $\mathbf{D}$ with $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$, and compute $\mathbf{P}^{-1}$. (b) Write down $\mathbf{D}^5$ and $\mathbf{D}^k$ for a general positive integer k.

Exercise 1.7 *(⋆ drill – transition matrices and the steady state).* A transition matrix has non-negative entries and *columns that sum to one*. (a) Which of these qualify?

$$\text{(i)}\ \begin{pmatrix} 0.6 & 0.1 \\ 0.4 & 0.9 \end{pmatrix}, \quad \text{(ii)}\ \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.6 \end{pmatrix}, \quad \text{(iii)}\ \begin{pmatrix} 0.3 & 0.7 \\ 0.7 & 0.3 \end{pmatrix}.$$

(b) Find the steady-state distribution of matrix (i): the eigenvector for $\lambda = 1$, scaled so its entries sum to one.

2 Tutorial problems

Exercise 2.1 *(⋆⋆ core – a full 2×2 eigensystem).* For $\mathbf{A} = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$ find the eigenvalues and an eigenvector for each, and verify the trace/determinant identities.

Exercise 2.2 *(▶ ⋆⋆ core – diagonalise and find $\mathbf{A}^k$).* Let $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$. (a) Diagonalise $\mathbf{A} = \mathbf{P}\mathbf{D}\mathbf{P}^{-1}$. (b) Hence derive a closed form for $\mathbf{A}^k$, valid for every positive integer k. (c) Check your formula at $k = 1$.

Exercise 2.3 *(▶ ⋆⋆ core – a Binet-type formula).* A sequence satisfies $x_{n+1} = x_n + 2x_{n-1}$ with $x_0 = 0$, $x_1 = 1$ (giving $0, 1, 1, 3, 5, 11, \ldots$). (a) Write the step as $\binom{x_{n+1}}{x_n} = \mathbf{A}\binom{x_n}{x_{n-1}}$ and find $\mathbf{A}$ and its eigenvalues. (b) Decompose the start vector in the eigenbasis and obtain a closed form for x_n.

Exercise 2.4 *(▶ ⋆⋆ core – a Hermitian 2×2 (physics)).* Let $\mathbf{H} = \begin{pmatrix} 3 & 2\mathrm{i} \\ -2\mathrm{i} & 3 \end{pmatrix}$. (a) Confirm $\mathbf{H}$ is Hermitian. (b) Find its eigenvalues (they should be real) and an eigenvector for each. (c) Verify the two eigenvectors are orthogonal under the conjugate inner product $\mathbf{u}^\dagger \mathbf{v} = \overline{u_1} v_1 + \overline{u_2} v_2$.

Exercise 2.5 *(▶ ⋆⋆ core – orthogonal diagonalisation of a symmetric 3×3).* The symmetric matrix $\mathbf{A} = \begin{pmatrix} 3 & 1 & -1 \\ 1 & 3 & -1 \\ -1 & -1 & 5 \end{pmatrix}$ has eigenvalues $\lambda = 2, 3, 6$, with eigenvectors $(1, -1, 0)^{\mathsf{T}}$, $(1, 1, 1)^{\mathsf{T}}$, $(1, 1, -2)^{\mathsf{T}}$ respectively. (a) Verify one eigenpair directly, say $\lambda = 6$. (b) Confirm the three eigenvectors are mutually orthogonal. (c) Normalise them to build an orthogonal matrix $\mathbf{Q}$ and write the spectral decomposition $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}$.

Exercise 2.6 *(⋆⋆ core – a defective matrix).* Show that $\mathbf{A} = \begin{pmatrix} 5 & -1 \\ 1 & 3 \end{pmatrix}$ is *not* diagonalisable.

Exercise 2.7 *(▶ ⋆⋆ core – a two-state Markov chain (data science)).* A streaming service tracks subscribers. Each month a *subscribed* user stays subscribed with probability 0.8 and lapses with 0.2; a *lapsed* user resubscribes with 0.4 and stays lapsed with 0.6. Let $\mathbf{x}_n$ = (subscribed, lapsed) fractions. (a) Write the transition matrix $\mathbf{P}$ and find its eigenvalues; (b) find the steady-state distribution; (c) starting from everyone subscribed, $\mathbf{x}_0 = (1, 0)$, find a closed form for $\mathbf{x}_n$ and state the long-run subscribed fraction.

Exercise 2.8 *(⋆⋆ core – a spiral orbit (physics)).* Let $\mathbf{A} = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}$ and $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$. (a) Find the eigenvalues of $\mathbf{A}$ and an eigenvector. (b) Starting from $\mathbf{x}_0 = (1, 0)^{\mathsf{T}}$, compute $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$, plot the four points and sketch the orbit through them. (c) Relate what you draw to the modulus and argument of the eigenvalues.

Exercise 2.9 *(⋆⋆ core – true or false).* State whether each is true or false, with a one-line justification.

(a) Every real 2×2 matrix has at least one real eigenvalue.

(b) If $\mathbf{A}$ is invertible, 0 is not an eigenvalue of $\mathbf{A}$.

(c) A 3×3 matrix with eigenvalues $2, 2, 5$ can never be diagonalised.

(d) Eigenvectors belonging to distinct eigenvalues are linearly independent.

(e) A real symmetric matrix can have a non-real eigenvalue.

(f) If λ is an eigenvalue of $\mathbf{A}$, then λ^2 is an eigenvalue of $\mathbf{A}^2$.

Exercise 2.10 *(▶ ⋆⋆⋆ challenge – capstone proof: symmetric ⇒ real)*. Show that a real symmetric matrix $\mathbf{A} = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$ always has real eigenvalues. (This is the first half of the 2×2 spectral theorem that underpins the orthogonal diagonalisation in Exercise 2.5.)

3 Further practice (home)

Exercise 3.1 *(⋆⋆ core – two 2×2 eigensystems)*. For each matrix find the eigenvalues and an eigenvector for each (allow complex values, and note any defective case):

$$\text{(a)} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \qquad \text{(b)} \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix}.$$

Exercise 3.2 *(⋆⋆ core – a 3×3 with a convenient zero)*. Find all eigenvalues and an eigenvector for each of $\mathbf{A} = \begin{pmatrix} 2 & 3 & 0 \\ 3 & 3 & 3 \\ -1 & 0 & 2 \end{pmatrix}$, and check your eigenvalues against $\operatorname{tr}\mathbf{A}$ and $\det\mathbf{A}$.

Exercise 3.3 *(⋆⋆ core – another matrix power)*. Let $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 2 & 1 \end{pmatrix}$. Diagonalise it and find a closed form for $\mathbf{A}^n$.

Exercise 3.4 *(⋆⋆ core – another Hermitian matrix)*. Let $\mathbf{H} = \begin{pmatrix} 4 & 1+\mathrm{i} \\ 1-\mathrm{i} & 3 \end{pmatrix}$. Check it is Hermitian, find its eigenvalues and eigenvectors, and verify the eigenvectors are orthogonal under the conjugate inner product. What would the plain product $\mathbf{u}^{\mathsf{T}}\mathbf{v}$ have said?

Exercise 3.5 *(⋆⋆ core – algebraic versus geometric multiplicity)*. For each matrix, give the algebraic and geometric multiplicity of every eigenvalue, and say whether it is diagonalisable:

$$\text{(a)} \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad \text{(b)} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \qquad \text{(c)} \begin{pmatrix} 4 & 2 \\ 0 & 1 \end{pmatrix}.$$

Exercise 3.6 *(⋆⋆ core – a covariance matrix (data science))*. A data set has (scaled) covariance matrix $\mathbf{A} = \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix}$. (a) Find its eigenvalues and eigenvectors. (b) Confirm the eigenvectors are orthogonal. (c) In PCA terms, along which direction does the data vary, and how much variance lies in the perpendicular direction?

Exercise 3.7 *(⋆⋆ core – spectral decomposition as PCA)*. The symmetric matrix $\mathbf{A} = \begin{pmatrix} 3 & 2 & 0 \\ 2 & 4 & 2 \\ 0 & 2 & 5 \end{pmatrix}$ (think of it as a covariance matrix) has eigenvalues $\lambda = 1, 4, 7$, with eigenvectors $(2, -2, 1)^{\mathsf{T}}$, $(2, 1, -2)^{\mathsf{T}}$, $(1, 2, 2)^{\mathsf{T}}$ respectively. (As in Exercise 2.5, the eigenvectors are supplied so the work stays clean.) (a) Verify one eigenpair directly, say $\lambda = 7$. (b) Confirm the three eigenvectors are mutually orthogonal. (c) Normalise them to build an orthogonal matrix $\mathbf{Q}$ and write the spectral decomposition $\mathbf{A} = \mathbf{Q}\mathbf{D}\mathbf{Q}^{\mathsf{T}}$. (d) In PCA language, which direction is the first principal axis and what fraction of the total variance does it carry?

Exercise 3.8 *(⋆ ⋆ ⋆ challenge – Lucas numbers and the golden ratio)*. The Lucas numbers obey $L_{n+1} = L_n + L_{n-1}$ with $L_0 = 2$, $L_1 = 1$ $(2, 1, 3, 4, 7, 11, \dots)$. Using the matrix $\mathbf{A} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with eigenvalues the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$ and $\psi = \frac{1-\sqrt{5}}{2}$ (eigenvectors $(\varphi, 1)^{\mathsf{T}}, (\psi, 1)^{\mathsf{T}}$), show that $L_n = \varphi^n + \psi^n$.

Exercise 3.9 *(⋆ ⋆ ⋆ challenge – when does a chain settle?)*. A Markov chain settles to a unique steady state, whatever its start, if $\lambda = 1$ occurs just once and every other eigenvalue has modulus less than 1. For each transition matrix, find the eigenvalues and say whether the chain settles or keeps oscillating for ever:

$$\text{(a) } \mathbf{P} = \begin{pmatrix} 0.2 & 0.6 \\ 0.8 & 0.4 \end{pmatrix}, \qquad \text{(b) } \mathbf{P} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

(c) For the chain in (a), starting from $\mathbf{x}_0 = (1, 0)$, find a closed form for $\mathbf{x}_n$ and its limit, and describe how $\mathbf{x}_n$ approaches that limit.